

ARDUINO PHYSICAL AI CHALLENGE INDIA 2026

Project Report

Project Title	TerraCast AI – Weather-Aware Smart Irrigation for Home Plants
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1. Project Overview

Problem Statement

Home gardeners and small growers can over-water plants when irrigation follows a fixed schedule. This wastes water and can keep soil excessively wet, while watering during or immediately before rain can be unnecessary. TerraCast AI addresses this by combining real-time soil-moisture measurements with weather information to make an automatic watering decision for a demonstration plant.

How Your Project Works

TerraCast AI uses an Arduino UNO Q as the primary control board. A soil-moisture sensor is sampled through A0 and converted into a moisture percentage and soil state. The current calibration defines RAW values below 1000 as WET, 1000–1200 as MODERATE, and values above 1200 as DRY. Weather information is supplied to the UNO Q software side, including temperature and rain probability. In LIVE mode, watering is allowed only when the soil is dry and rain probability is below 60%. Once the pump starts, it remains on until the sensor reaches the wet threshold of $RAW \leq 1000$, subject to a two-minute safety timeout. An L298N drives the DC pump. A 128×64 I2C OLED displays city, temperature, moisture, raw sensor value, soil state, weather mode and watering status. The built-in LED matrix provides a compact visual state display, while a push button cycles through SUNNY, RAIN, WATERING, PLANT and LIVE demonstration modes.

Why Arduino UNO Q?

The UNO Q is the primary board because it combines Arduino real-time control with an onboard Linux-capable software environment. This makes it practical to read the sensor, control the pump and displays, while also handling weather-data processing and communication between the software and hardware sides. The final prototype uses this capability for weather-aware coordination; it does not claim a trained machine-learning model in the current version.

2. Components Used (BOM)

Component	Qty
Arduino UNO Q (ABX00087)	1
128×64 SSD1306 I2C OLED	1
Soil moisture sensor	1
L298N motor driver	1
DC water pump	1
Demo push button	1
External pump power source / battery pack	1
Jumper wires / breadboard wiring	As required

3. System Architecture & Circuit

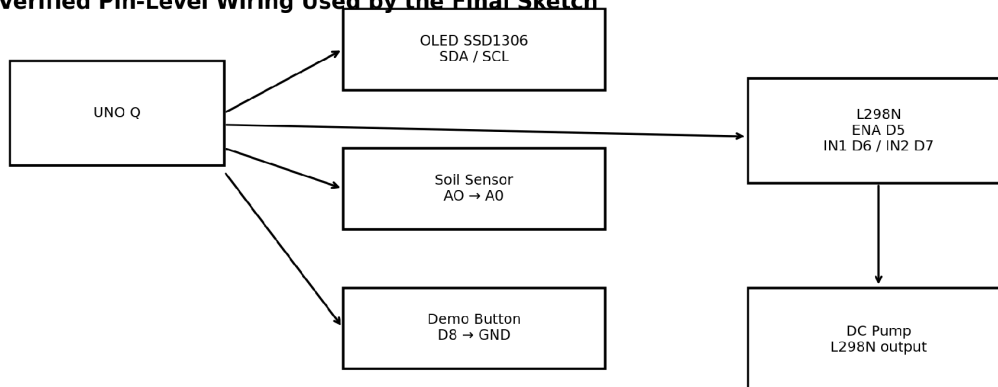
Step-by-Step Workflow

1. Read and average soil-moisture samples from A0.
2. Convert the raw reading into a moisture percentage and classify the soil as WET, MODERATE or DRY.
3. Receive/update weather values such as temperature and rain probability.
4. In LIVE mode, if soil RAW >1200 and rain probability <60%, start the pump through the L298N.
5. Keep the pump running while monitoring the soil.
6. Stop the pump when RAW ≤1000, when significant rain is expected, or when the two-minute safety timeout is reached.
7. Update the OLED and LED matrix so the physical state is visible.
8. Use the demo button to cycle through SUNNY, RAIN, WATERING, PLANT and LIVE.

Block Diagram & Circuit Schematic

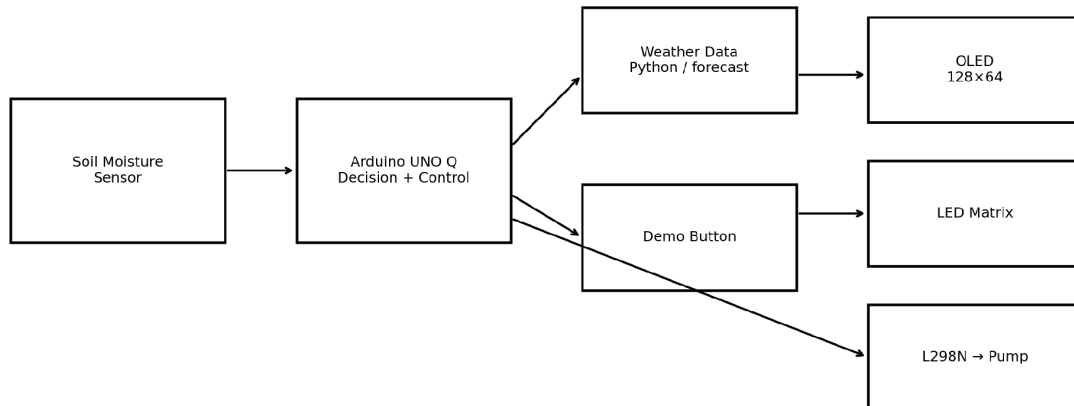
Circuit / Wiring Diagram

Verified Pin-Level Wiring Used by the Final Sketch



Common GND is required between UNO Q and L298N. Pump power is supplied through the L298N motor supply.

TerraCast AI System Architecture



Brief Description & Limitations

The current TerraCast AI prototype is a Physical AI system rather than a trained ML system. Its decision engine uses measured soil moisture and a weather rain-probability threshold to control a physical pump. This makes the behavior deterministic and easy to demonstrate: dry soil triggers irrigation only when rain probability is below 60%, and irrigation stops at $RAW \leq 1000$. Limitations include sensor-to-sensor calibration differences, dependence on the freshness and quality of weather data, and the absence of learning from historical plant data. A future version could use collected soil, weather and plant-growth data to train an irrigation predictor.

5. Code Structure

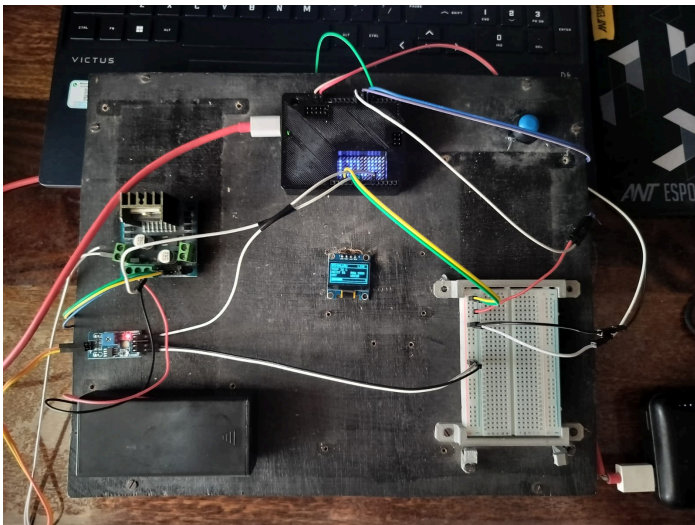
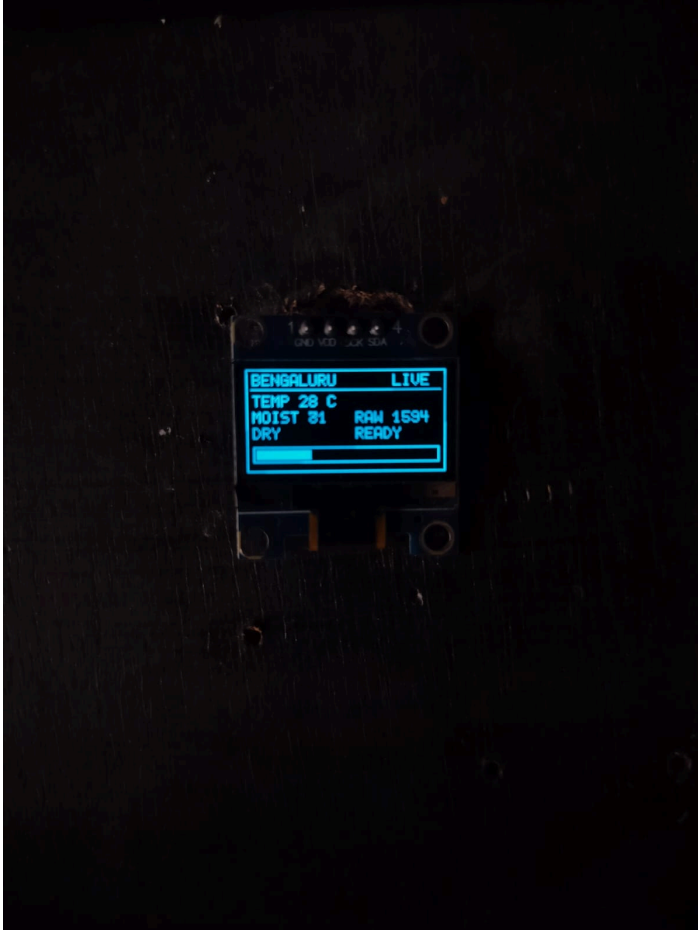
The Arduino sketch is organized into clear functional sections. `setup()` initializes the soil sensor, pump driver, demo button, LED matrix, OLED and UNO Q bridge. `loop()` handles button events, periodic soil sampling, automatic watering, pump monitoring, OLED refresh and matrix refresh. `readSoil()` averages sensor readings and calculates the moisture percentage. `automaticWatering()`, `startPump()`, `stopPump()` and `updatePump()` implement the irrigation state machine and safety rules. `setForecast()` receives weather values from the software side. `drawOLED()` renders the dashboard, while `updateMatrix()` selects the current visual state. The demo button cycles through SUNNY → RAIN → WATERING → PLANT → LIVE → SUNNY.

GitHub Repository

<https://github.com/SHRIY8N/TerraCast-AI>

6. Testing & Results

Testing was performed by changing the soil sensor conditions and observing the complete sensor-to-actuator chain. The calibration points were verified as: RAW <1000 = WET, RAW 1000–1200 = MODERATE, and RAW >1200 = DRY. In the dry condition, the pump starts when rain probability is below 60% and remains on while the soil is monitored. The pump stops when RAW reaches 1000 or below. When rain probability is 60% or higher, automatic watering is blocked. A two-minute pump safety timeout prevents indefinite operation if the sensor fails to reach the wet threshold. The demo button was tested through all five states, including the fifth LIVE state so the system can return to real-time operation. The OLED and LED matrix were checked for each state. No numerical ML accuracy is claimed because the current prototype does not use a trained ML model.



7. Challenges, Learnings & Future Improvements

Challenges Faced

- Soil sensor calibration: raw ADC values depend on the soil, sensor placement and moisture distribution, so project-specific thresholds were used.
 - OLED rendering: the display required a direct SSD1306 I2C implementation, and numeric glyphs were separated from the letter font to prevent mirrored/reversed digits.
 - LED matrix graphics: the supplied reference icons were converted into compact 12×8 patterns for the UNO Q matrix.
 - Irrigation logic: the pump needed to remain on until the soil reached the wet threshold instead of stopping after a short fixed interval.
 - Power and motor control: the L298N is used as the pump driver, with common ground between the UNO Q control system and motor driver.
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What You Learned & What's Next

The project demonstrated how sensing, software decisions and physical actuation can be combined into a complete Physical AI-style prototype. The build also highlighted the importance of sensor calibration, electrical grounding, motor-driver isolation from logic power, robust display rendering and safety conditions in actuator control.

8. Declaration

I confirm this is our original, unpublished work. The Arduino® UNO™ Q is the primary board in this project.

Date	23 August 2026
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